

CS112: Monday 31 August, 12:15 to 1:20

DRAFT, RECONSTRUCTED FROM COURSE-MAP.YML AND THE TEACHING LOG

You arrive from Python, where running a program is one step. You wrote the file, you typed `python thing.py`, and it ran. Nothing stood between the text you wrote and the thing that happened.

Today: your source file is not the program. Something has to turn it into one, and that something is a chain of tools with three separate places to go wrong. Knowing which stage failed is most of what reading a C++ error message is.

TIME	BEAT
12:20	1. The shape of a C++ program
12:32	2. The pipeline, and where it breaks
12:50	3. Errors, live

Beat 1. The shape of a C++ program (12 min)

```
#include <iostream>
using namespace std;

int main() {
    cout << "hello" << endl;
    return 0;
}
```

Four things, named while typing it: `#include` pulls in a file before compilation begins; `main` is where the program starts and it returns an `int`; `cout <<` sends a value out, `cin >>` reads one in; `return 0` is how a program reports that nothing went wrong.

Talk to your neighbor. What does the compiler actually do to this file?

Beat 2. The pipeline, and where it breaks (18 min)

```
hello.cpp --preprocess--> --compile--> hello.o --link-->
hello
```

Each arrow can fail, and they fail differently.

- A **compile error** means one file does not make sense on its own: a missing semicolon, a name never declared, a type mismatch. The compiler names the file and the line.
- A **link error** means every file made sense alone and together they are missing something: a function declared and never written. There is no line number, because there is no line; the thing is simply not there.

`make` exists for that distinction. It knows which files changed and which step each one needs next.

Beat 3. Errors, live (15 min)

Break the program on purpose and read what comes back, one at a time. Delete a semicolon. Misspell `cout`. Declare a function and never define it.

Which stage failed, and how can you tell from the message alone?

The habit that starts today: **read the first error, not the last one.** One missing semicolon produces a cascade, and every message after the first is the compiler guessing about wreckage it caused itself.